

DSC602 INFORMATION VISUALIZATION - COMPLETE EXAM STUDY GUIDE

University of Buea | Faculty of Science | Computer Science

EXAM PAPER OVERVIEW

You have been given **2 past exam papers** for DSC602:

Aspect	June 2025 (Theory)	June 2024 (Practical)
Total Marks	70	15
Duration	3 hours	2 hours
Format	Written exam	Jupyter Notebook (Lab)
Focus	Theory-heavy	Practical-heavy
Questions	3 major sections	5 integrated questions
Data	Conceptual	Health dataset (Fitness tracking)

QUICK COMPARISON: JUNE 2025 vs JUNE 2024

June 2025 Exam (70 marks, 3 hours)

Question 1: Reading & Comprehension (25 marks)

- Q1.1: Visualization types definition (6)
- Q1.2: Business need for visualization (6)
- Q1.3: Governance applications (6)
- Q1.4: Three graphics from class (6)
- Mark allocation: Theory-focused

Question 2: Data Description & Preparation (20 marks)

- Q2.1: Dictionary, Tuple, Series (9)
- Q2.2: Obtain/Maintain/Use data structures (6)
- Q2.3: DataFrame definition & particularity (5)
- Mark allocation: Theory + basic code examples

Question 3: Graphics & Characteristics (25 marks)

- 6 graph types: Line, Bar, Histogram, Flow, Box, Bubble
 - For each: Definition + Best contexts + Justification
 - Mark allocation: 4 marks × 6 graphs + 1 bonus
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June 2024 Exam (15 marks, 2 hours - Lab-based Jupyter)

Question 1: InfoViz Definition (3 marks)

- Simple definition question
- Understanding core concepts

Question 2: Python Environment (3 marks)

- Libraries needed
- Installation commands
- Minimum setup

Question 3: Correlation Matrix (4 marks)

- What is it (1)
- Why necessary (1)
- Generate matrix (1)
- Analyze results (1)

Question 4: Scatter Plots (3 marks)

- Create visualizations from correlation insights
- Multiple plot types
- Statistical validation

Question 5: Multivariate Regression (3 marks)

- Build regression model
- Target: Hours_Sleep
- Predict and analyze

KEY CONCEPTS MASTERY

CONCEPT 1: Information Visualization vs Data Visualization

Information Visualization (InfoViz):

- Abstract, non-spatial data → visual form
- Maps concepts to graphics
- Examples: Org charts, network diagrams, timelines
- Requires transformation step

Data Visualization:

- Inherently spatial data
- Geographic context preserved
- Examples: Maps, satellite imagery
- Minimal transformation needed

Key Difference: InfoViz creates visual representations where none existed; Data Viz displays existing spatial information.

CONCEPT 2: Data Structures for Visualization

Dictionary

```
config = {'plot_type': 'bar', 'colors': ['red', 'blue']}  
# Unordered, mutable, access by key  
# Use: Store configuration parameters
```

Tuple

```
point = (10, 20) # Coordinates  
colors = (255, 128, 0) # RGB  
# Ordered, immutable, indexed access  
# Use: Ensure data integrity
```

Series (Pandas)

```
sales = pd.Series([100, 150, 120], index=['Q1', 'Q2', 'Q3'])  
# Labeled, 1D, optimized for data ops  
# Use: Direct plotting, time series
```

DataFrame

```
df = pd.DataFrame({'Name': ['A', 'B'], 'Sales': [100, 200]})  
# 2D, labeled, heterogeneous, vectorized ops  
# Use: Core data structure for analysis
```

Why DataFrame Particularly Efficient:

1. **Vectorized Operations:** `df['Bonus'] = df['Salary'] * 0.1` (100x faster than loops)
 2. **Built-in Methods:** `df.groupby()`, `df.fillna()`, `df.corr()`
 3. **Automatic Indexing:** Semantic meaning in row/column labels
 4. **Operations Integration:** Filter, aggregate, join, reshape - all efficient
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CONCEPT 3: Graph Type Selection

Graph	Best For	Why Useful	Example
Line	Temporal trends	Pattern recognition, trends, seasonality	Stock prices over time
Bar	Category comparison	Precise magnitudes, simplicity	Sales by region
Histogram	Distribution analysis	Distribution shape, outliers	Age distribution
Flow Chart	Process/decisions	Clarity, branching logic	Customer service process
Box Plot	Statistical summary	Multi-group comparison, outliers	Salary by department
Bubble Map	Geographic analysis	Multi-dimensional, spatial patterns	Sales by location

CONCEPT 4: Correlation Matrix Interpretation

Correlation Strength:

- 0.9-1.0: Very strong
- 0.7-0.9: Strong
- 0.5-0.7: Moderate
- 0.3-0.5: Weak
- 0.0-0.3: Very weak/None

Using Correlation Matrix:

1. **Feature Selection:** Which variables predict the target?
2. **Multicollinearity:** Are predictors redundant?
3. **Pattern Recognition:** What hidden relationships exist?
4. **Model Building:** What features should we keep/remove?

Health Data Example:

- Duration ↔ Calories: $r = 0.98$ (multicollinearity - keep one only)
- Work_Hours ↔ Sleep: $r = -0.85$ (strong inverse - important predictor)
- Calories ↔ Sleep: $r = 0.62$ (moderate positive - fitness helps sleep)

CONCEPT 5: Regression Modeling

Simple vs Multivariate:

- **Simple:** One predictor → One line
- **Multivariate:** Multiple predictors → Better predictions

Regression Equation:

$$\text{Target} = \text{Intercept} + (\text{Coef}_1 \times \text{Var}_1) + (\text{Coef}_2 \times \text{Var}_2) + \dots$$

Model Evaluation:

- **R²**: Explains what % of variance (0.75 = good, 0.9 = excellent)
- **RMSE**: Average prediction error (lower is better)
- **MAE**: Absolute average deviation (interpretable in original units)

From June 2024 Exam:

$$\text{Hours_Sleep} = 9.2 - 0.18 \times \text{Hours_Work} + 0.008 \times \text{Calories} + \dots$$

- Interpretation: Each work hour → 0.18 hours (11 minutes) less sleep
- Important insight: Work-life balance impacts sleep significantly

PYTHON CODE PATTERNS

Pattern 1: Load & Explore Data

```
import pandas as pd
import numpy as np

df = pd.read_csv('health.csv')
df.head()      # First few rows
df.describe()  # Statistics
df.dtypes      # Column types
```

Pattern 2: Calculate Correlations

```
import seaborn as sns
import matplotlib.pyplot as plt

corr_matrix = df.corr()
sns.heatmap(corr_matrix, annot=True, cmap='coolwarm')
plt.show()
```

Pattern 3: Create Scatter Plot

```
plt.scatter(df['X'], df['Y'], alpha=0.6, s=100)
z = np.polyfit(df['X'], df['Y'], 1)
p = np.poly1d(z)
```

```
plt.plot(df['X'], p(df['X']), "r--", label='Trend')
plt.show()
```

Pattern 4: Build Regression Model

```
from sklearn.linear_model import LinearRegression
from sklearn.metrics import r2_score

X = df.drop('Target', axis=1)
y = df['Target']

model = LinearRegression()
model.fit(X, y)
y_pred = model.predict(X)

print(f"R² = {r2_score(y, y_pred):.4f}")
```

Pattern 5: Create Multiple Visualizations

```
fig, axes = plt.subplots(2, 2, figsize=(14, 10))

# Plot 1
axes[0, 0].scatter(...)

# Plot 2
axes[0, 1].plot(...)

# Plot 3
axes[1, 0].hist(...)

# Plot 4
axes[1, 1].bar(...)

plt.tight_layout()
plt.show()
```

EXAM STRATEGY & TIME MANAGEMENT

Before Exam

- ☐ Memorize 6 graph types with definitions & use cases
- ☐ Practice correlation matrix interpretation
- ☐ Understand regression coefficient interpretation
- ☐ Know Python library minimum requirements
- ☐ Review June 2024 practical (6-8 mark questions possible)
- ☐ Practice explaining concepts simply (non-technical audience)

During Exam

June 2025 Style (Written, 3 hours):

- **First 5 min:** Read all questions, identify easy vs hard
- **Q1 (45 min):** Theory answers - can do partial credit for partial explanations
- **Q2 (30 min):** Data structure definitions - use simple Python examples
- **Q3 (45 min):** For each graph: 1-2 min definition, 2 min best uses, 1 min justification
- **Review (5 min):** Check completeness, add examples if time

June 2024 Style (Practical, 2 hours):

- **First 5 min:** Understand health data structure
- **Q1-Q2 (10 min):** Quick definition answers
- **Q3 (30 min):** Generate correlation matrix, interpret correlations
- **Q4 (20 min):** Create 3 scatter plots with trend lines
- **Q5 (40 min):** Build regression, analyze coefficients, create diagnostic plots
- **Final (5 min):** Verify all code runs, exports cleanly

COMMON MISTAKES TO AVOID

Mistake 1: Dictionary vs Tuple Confusion

✗ "Dictionaries are ordered arrays"

✓ "Dictionaries are unordered key-value pairs; Tuples are ordered and immutable"

Mistake 2: Wrong Graph for Context

✗ Using pie charts (pie charts are rarely best choice)

✓ Using bar chart for categorical comparison

Mistake 3: No Justification

✗ "Bar charts are good for data"

✓ "Bar charts enable precise height comparison for 5 regions; alternative line graph would create visual clutter"

Mistake 4: Correlation = Causation

✗ "Since Work and Sleep correlate -0.85, work causes poor sleep"

✓ "Correlation -0.85 shows inverse relationship; causation unclear (could be stress, could be schedule)"

Mistake 5: Ignoring Multicollinearity

✗ Using Duration + Average_Pulse + Max_Pulse (all $r > 0.95$)

✓ Using Duration only (removes redundancy)

Mistake 6: Not Interpreting R^2

✗ Just printing " $R^2 = 0.76$ "

✓ "Model explains 76% of variance; 24% due to other factors (stress, schedule, genetics)"

MARK ALLOCATION CHECKLIST

June 2025 (70 marks)

Q1: Reading & Comprehension (25 marks)

- ☐ Q1.1a: InfoViz definition (3)
- ☐ Q1.1b: Modern context (3)
- ☐ Q1.2: 6 business reasons (6)
- ☐ Q1.3: 6 governance examples (6)
- ☐ Q1.4: 3 graphs with explanation (6)

Q2: Data Structures (20 marks)

- ☐ Q2.1: Dict+Tuple+Series definitions (9)
- ☐ Q2.2: Obtain/Maintain/Use (6)
- ☐ Q2.3: DataFrame definition + efficiency (5)

Q3: Graphics (25 marks)

- ☐ Graph 1 (Line): 4
- ☐ Graph 2 (Bar): 4
- ☐ Graph 3 (Histogram): 4
- ☐ Graph 4 (Flow): 4
- ☐ Graph 5 (Box): 4
- ☐ Graph 6 (Bubble): 4
- ☐ Bonus: 1

TOTAL: 70 marks

June 2024 (15 marks)

- ☐ Q1: Definition (3)
- ☐ Q2: Python environment (3)
- ☐ Q3: Correlation matrix (4)
- ☐ Q4: Scatter plots (3)
- ☐ Q5: Regression model (2-3)

TOTAL: 15 marks

STUDY RESOURCES PROVIDED

Markdown Files (Theory):

1. [DSC602_JUNE_2025_EXAM_SOLUTIONS.md](#) - Complete theory answers (70 marks)

2. [DSC602_JUNE_2024_EXAM_SOLUTIONS.md](#) - Complete theory answers (15 marks)
3. [DSC602_JUNE_2025_QUICK_REFERENCE.md](#) - Study guide for June 2025
4. [DSC602_JUNE_2024_EXAM_QUICK_REFERENCE.md](#) - Study guide for June 2024

Jupyter Notebooks (Interactive Practice):

1. [DSC602_JUNE_2025_TITANIC_Q3_VISUALIZATIONS.ipynb](#) - All 6 graph types (run cells to learn)
2. [DSC602_JUNE_2024_EXAM_PRACTICAL.ipynb](#) - Full practical exam walkthrough

Python Scripts (Standalone Execution):

1. [DSC602_JUNE_2025_Q3_VISUALIZATIONS.py](#) - Run: `python DSC602_JUNE_2025_Q3_VISUALIZATIONS.py`
2. [DSC602_JUNE_2024_EXAM.py](#) - Run: `python DSC602_JUNE_2024_EXAM.py`

30-MINUTE CRASH COURSE

If you have 30 minutes before exam:

5 min: Graph types

- Line = Trends, Bar = Categories, Histogram = Distribution, Flow = Process, Box = Multi-group, Bubble = Geographic

5 min: Data structures

- Dictionary: key-value (flexible)
- Tuple: immutable (fixed)
- Series: labeled 1D (for plotting)
- DataFrame: 2D labeled (core analytics)

5 min: Python basics

- `import pandas as pd, import matplotlib.pyplot as plt, import seaborn as sns`
- `df.corr()` for correlations
- `LinearRegression().fit(X, y)` for modeling

5 min: Correlation interpretation

- $r > 0.9$: Very strong (multicollinearity)
- $r = 0.6$: Moderate (useful predictor)
- $r = -0.85$: Strong negative (inverse relationship)

5 min: Regression interpretation

- R^2 = what % variance explained
- Coefficients: change per unit increase
- RMSE: average prediction error

5 min: Last minute prayers 🙏

FINAL TIPS

✅ DO THIS

1. **Be specific:** Don't say "visualization helps" - say "bar charts enable 3-second comparison of 5 regions"
2. **Use examples:** Abstract + concrete = highest marks
3. **Show working:** Partial credit for correct method even if calculation wrong
4. **Organize:** Bullet points, clear sections, bold key terms
5. **Proofread:** 2-3 min check at end catches silly errors

❌ AVOID THIS

1. **Generic definitions:** "A line graph shows lines" (too vague)
2. **Missing justification:** Why use this graph? For whom?
3. **Confusing structure:** Does your answer flow logically?
4. **No code examples:** For data structures, simple Python code helps
5. **Rushing:** Slower, complete answer > faster, incomplete answer

YOUR EXAM FILES

```
d:\charlesDevelopment\SQL\note_summaries\DSC602\  
├─ DSC602_JUNE_2025_EXAM_SOLUTIONS.md  
├─ DSC602_JUNE_2025_QUICK_REFERENCE.md  
├─ DSC602_JUNE_2025_TITANIC_Q3_VISUALIZATIONS.ipynb  
├─ DSC602_JUNE_2025_Q3_VISUALIZATIONS.py  
├─  
├─ DSC602_JUNE_2024_EXAM_SOLUTIONS.md  
├─ DSC602_JUNE_2024_EXAM_PRACTICAL.ipynb  
├─ DSC602_JUNE_2024_EXAM.py  
├─  
└─ (This file)
```

How to use:

1. **Study:** Read **.md** files first (understand concepts)
2. **Practice:** Run **.ipynb** files in Jupyter (interactive learning)
3. **Execute:** Run **.py** scripts (see complete output)
4. **Review:** Use quick reference guides before exam

FINAL CHECKLIST

Before June 2025 Exam (Written, 70 marks):

- ☐ Can define all 6 graph types without notes
- ☐ Can explain why each graph type is useful
- ☐ Can describe 3 governance use cases
- ☐ Understand difference: Dictionary vs Tuple vs Series

- ☐ Know why DataFrame is "particularly efficient"
- ☐ Can interpret correlation coefficients
- ☐ Managed timing: 45+30+45 min = 120 min (1 hr buffer)

Before June 2024 Exam (Practical, 15 marks):

- ☐ Can generate correlation matrix in Python
- ☐ Can create scatter plots with trend lines
- ☐ Can build LinearRegression model
- ☐ Can interpret R^2 , RMSE, coefficients
- ☐ Can create diagnostic plots (4-subplot layout)
- ☐ All code runs without errors (tested)

GOOD LUCK! You've thoroughly prepared. 📊🌟

Remember: The examiners want to see that you **understand concepts** and can **apply them to real situations**. Depth beats breadth. A detailed, well-explained answer to one question beats brief answers to everything.

You've got this! 💪